Physical Chemistry CUME

April 9, 2010

1) a) In 1920 Weigert published a seminal paper stating the following " The degree of the polarization increases with increasing molecular size, with increasing viscosity of the medium and with decreasing temperature"
Explain the basis for this statement **(15 points)**

b) Using your explanation, explain why Stokes had previously found in 1852 that quinine dissolved in water at room temperature showed no apparent polarization in its emission. **(10 points)**

2) a) What are the theoretical limitations for (i) polarization? and (ii) anisotropy? **(15 pts)**
b) Explain why these theoretical limitations cannot be experimentally realized in solution. **(15 pts)**

3) Describe why many fluorescent probes show a wavelength dependence on polarization in **highly viscous** solvents – the so-called Excitation Polarization Spectrum – but that almost all fluorescent probes show no wavelength dependence on emission wavelength in **highly viscous** solvents? **(20 pts)**

4) (a) Explain the observation by Gaviola and Pringsheim in 1924 that the polarization of fluorescein in **viscous** solvents decreased as the concentration of the dye increased (at a fixed solvent viscosity)? **(10 pts)**.
(b) Describe an experiment in which this type of effect was later taken advantage of to monitor protease activity? **(15 pts).**